

On the Computation of the Cosine Measure in High Dimensions

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Outline

- 1 Introduction
- 2 Reformulation
- 3 Algorithms
- 4 Test Set Construction
- 5 Numerical Results

Positive span of a set

- **Definition.** Let $S \subseteq \mathbb{R}^n$. The positive span of S is the set of all non-negative linear combinations of vectors in S :

$$\text{pspan}(S) = \left\{ \sum_{i=1}^m \alpha_i d_i : \alpha_i \geq 0, d_i \in S, m \in \mathbb{N} \right\}.$$

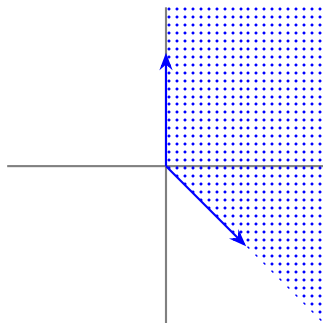
The set S is a positive spanning set for $\mathbb{R}^n$ if $\text{pspan}(S) = \mathbb{R}^n$.

- Many blackbox optimization algorithms require the set of search direction vectors to be positive spanning to ensure convergence.

Positive span of a set

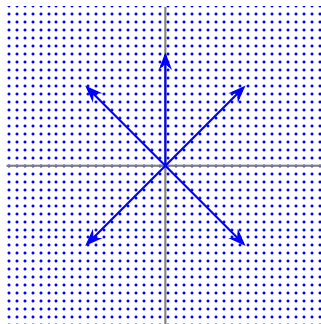
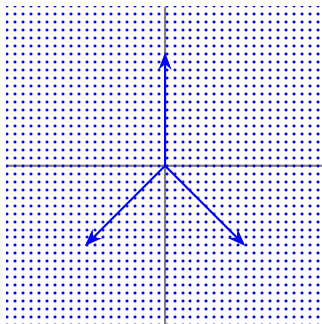
Recall: $\text{pspan}(S) = \{\sum_{i=1}^m \alpha_i d_i : \alpha_i \geq 0, d_i \in S, m \in \mathbb{N}\}.$

$$S = \left\{ (0, 1), \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right) \right\}$$



The set S does not positively span $\mathbb{R}^2$.

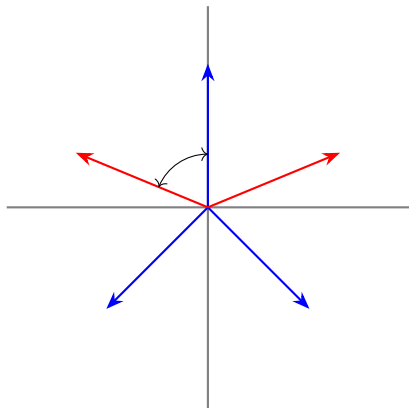
Positive span of a set



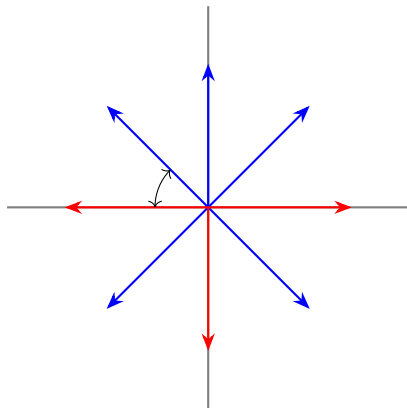
Both sets are positive spanning, however the example on the right expresses a richer set of possible search directions. How can we quantify this?

The cosine measure intuition

Let $S = \{d_1, \dots, d_k\} \subset \mathbb{R}^n$ be nonempty, finite, unit vectors.



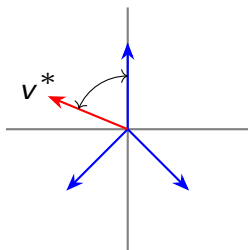
$$\cos\left(\frac{3\pi}{8}\right) \approx 0.383$$



$$\cos\left(\frac{\pi}{4}\right) \approx 0.707$$

Terminology related to the cosine measure

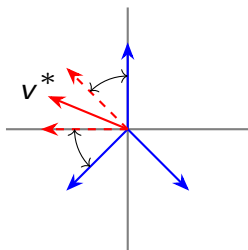
- The **cosine measure of S with respect to the reference vector v** denoted by $\text{cm}(S, v)$ is the cosine of the smallest angle between v and any element of S .



$$\text{cm}(S, v^*) = \cos\left(\frac{3\pi}{8}\right) \approx 0.383$$

Terminology related to the cosine measure

- The **cosine measure of S with respect to the reference vector v** denoted by $\text{cm}(S, v)$ is the cosine of the smallest angle between v and any element of S .
- A vector v belongs to the **cosine vector set** denoted by $\text{cV}(S)$ if v makes the largest possible angle with S .



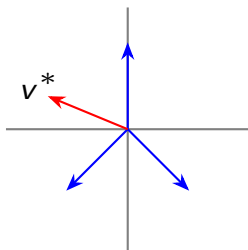
$$\text{cm}(S, v^*) = \cos\left(\frac{3\pi}{8}\right) \approx 0.383$$

$$v^* \in \text{cV}(S)$$

Terminology related to the cosine measure

- The **cosine measure of S with respect to the reference vector v** denoted by $\text{cm}(S, v)$ is the cosine of the smallest angle between v and any element of S .
- A vector v belongs to the **cosine vector set** denoted by $\text{cV}(S)$ if v makes the largest possible angle with S .
- The **cosine measure of S** denoted by $\text{cm}(S)$ is the cosine of the largest angle between some v and S . That is

$$\text{cm}(S) = \text{cm}(S, v) \text{ for } v \in \text{cV}(S).$$



$$\text{cm}(S, v^*) = \cos\left(\frac{3\pi}{8}\right) \approx 0.383$$

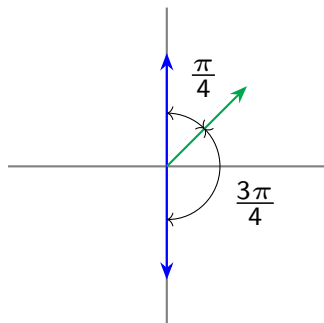
$$v^* \in \text{cV}(S)$$

$$\text{cm}(S) = \text{cm}(S, v^*) \approx 0.383$$

Cosine measure with respect to a reference vector

- Recall $a^\top b = \|a\| \|b\| \cos(\theta)$.
- Definition.** The cosine measure with respect to a reference vector $v \in \mathbb{R}^n$ is the cosine of the smallest angle between v and S is given by

$$\text{cm}(S, v) = \max_{1 \leq i \leq k} d_i^\top v.$$

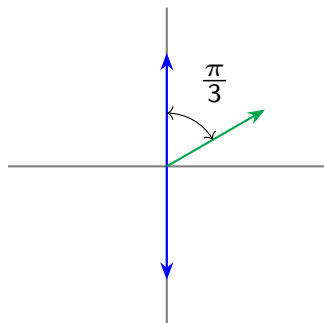


$$\begin{aligned}\text{cm}(S, v) &= \max \{e_2^\top v, -e_2^\top v\} \\ &= \max \left\{ \cos\left(\frac{\pi}{4}\right), \cos\left(\frac{3\pi}{4}\right) \right\} \\ &= \cos\left(\frac{\pi}{4}\right) \approx 0.707\end{aligned}$$

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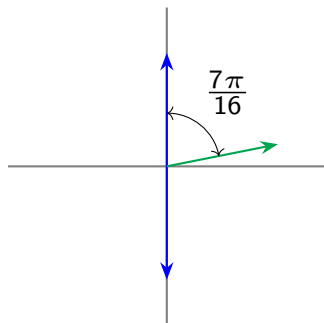


$$\begin{aligned}\text{cm}(S, v) &= \max \{e_2^\top v, -e_2^\top v\} \\ &= \max \left\{ \cos\left(\frac{\pi}{3}\right), \cos\left(\frac{2\pi}{3}\right) \right\} \\ &= \cos\left(\frac{\pi}{3}\right) = 0.5\end{aligned}$$

Cosine measure with respect to a reference vector

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- Definition.** The cosine measure with respect to a reference vector $v \in \mathbb{R}^n$ is the cosine of the smallest angle between v and S is given by

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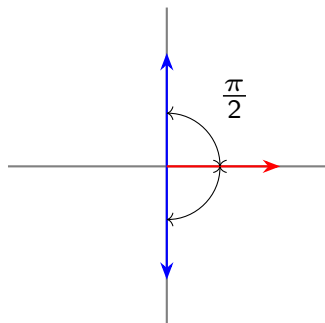


$$\begin{aligned}\text{cm}(S, v) &= \max \left\{ e_2^\top v, -e_2^\top v \right\} \\ &= \max \left\{ \cos\left(\frac{7\pi}{16}\right), \cos\left(\frac{9\pi}{16}\right) \right\} \\ &= \cos\left(\frac{7\pi}{16}\right) \approx 0.195\end{aligned}$$

Cosine measure with respect to a reference vector

- Recall $a^\top b = \|a\| \|b\| \cos(\theta)$.
- Definition.** The cosine measure with respect to a reference vector $v \in \mathbb{R}^n$ is the cosine of the smallest angle between v and S is given by

$$\text{cm}(S, v) = \max_{1 \leq i \leq k} d_i^\top v.$$



$$\begin{aligned} \text{cm}(S, v) &= \max \{e_2^\top v, -e_2^\top v\} \\ &= \max \left\{ \cos\left(\frac{\pi}{2}\right), \cos\left(\frac{\pi}{2}\right) \right\} \\ &= \cos\left(\frac{\pi}{2}\right) = 0 \end{aligned}$$

Formalizing the cosine measure

- Minimizing $\text{cm}(S, v)$ over all v where $\|v\| = 1$ yields the cosine of the largest angle between v and S .
- **Definition.** The cosine measure of the set S is the solution to the cosine measure problem (CMP)

$$\text{cm}(S) = \min_{\substack{v \in \mathbb{R}^n \\ \|v\|=1}} \text{cm}(S, v) = \min_{\substack{v \in \mathbb{R}^n \\ \|v\|=1}} \max_{1 \leq i \leq k} d_i^\top v$$

and yields the cosine of the largest angle between a non-zero vector $v \in \mathbb{R}^n$ and S . The set of minimizers to this problem is $c\mathbf{V}(S)$.

- Goal: find solutions to this problem in higher dimensions. This problem is NP-Hard [JM20].

Gram matrices

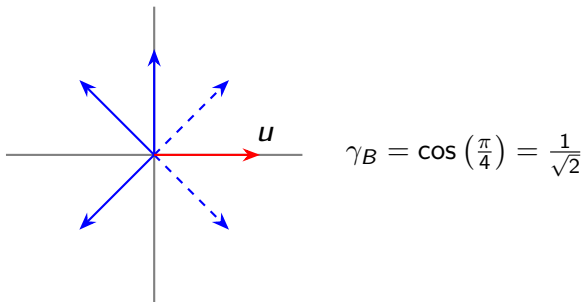
Definition. Let $S = [d_1 \ d_2 \ \dots \ d_k]$ be an $n \times k$ real matrix. The *Gram matrix* of S , denoted $\mathbf{G}(S)$, is defined to be

$$\mathbf{G}(S) = S^{\top} S = \begin{bmatrix} d_1^{\top} d_1 & d_1^{\top} d_2 & \cdots & d_1^{\top} d_k \\ d_2^{\top} d_1 & d_2^{\top} d_2 & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ d_k^{\top} d_1 & \cdots & \cdots & d_k^{\top} d_k \end{bmatrix}.$$

Gram matrices and the cosine measure

Theorem. Let $B = [d_1 \ d_2 \ \dots \ d_n] \in \mathbb{R}^{n \times n}$ where the columns are linearly independent unit vectors. Then there exists a unique unit vector $u \in \mathbb{R}^n$ such that $u^\top d_i = \gamma_B > 0$ for $i = 1, \dots, n$ where

$$\gamma_B = \frac{1}{\sqrt{e^\top \mathbf{G}(B)^{-1} e}}.$$

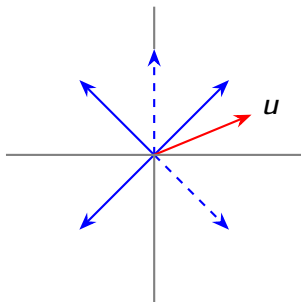


We call u the *Gram vector* and γ_B the *Gram value* of B .

Gram matrices and the cosine measure

Theorem. Let $B = [d_1 \ d_2 \ \dots \ d_n] \in \mathbb{R}^{n \times n}$ where the columns are linearly independent unit vectors. Then there exists a unique unit vector $u \in \mathbb{R}^n$ such that $u^\top d_i = \gamma_B > 0$ for $i = 1, \dots, n$ where

$$\gamma_B = \frac{1}{\sqrt{e^\top \mathbf{G}(B)^{-1} e}}.$$



$$\gamma_B = \cos\left(\frac{3\pi}{8}\right) \approx 0.383$$

We call u the *Gram vector* and γ_B the *Gram value* of B .

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Cosine measure as a non-convex QCLP

$$\begin{aligned} \min_{\substack{v \in \mathbb{R}^n \\ \|v\|=1}} \max_{1 \leq i \leq k} d_i^\top v & \iff \begin{aligned} & \min z \\ & \text{subject to: } d_i^\top v \leq z \\ & v^\top v = 1 \\ & v \in \mathbb{R}^n, z \in \mathbb{R} \end{aligned} \end{aligned}$$

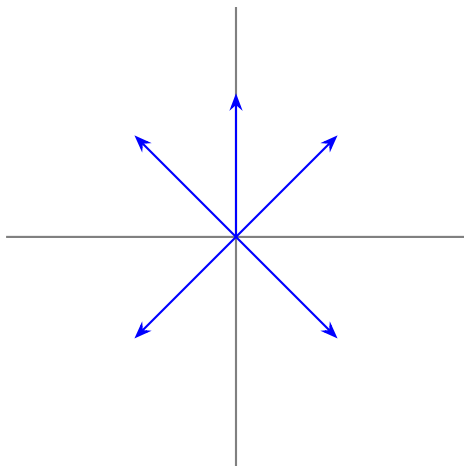
The solution $(v^*, z^*) \in \mathbb{R}^n \times \mathbb{R}$ gives the optimal cosine vector v^* and the cosine measure z^* .

Cosine measure as a non-convex QP

Theorem. The cosine measure problem is equivalent to the following norm maximization problem

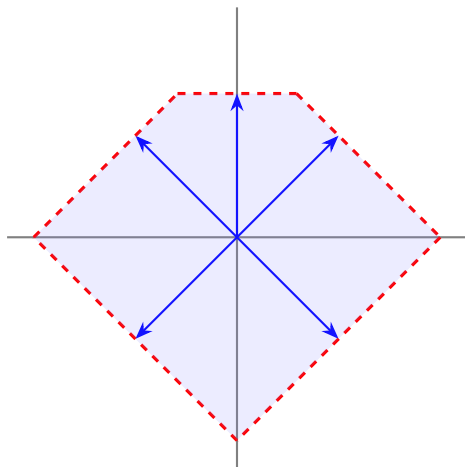
$$\begin{aligned} \max_{x \in \mathbb{R}^n} \quad & \|x\|^2 \\ \text{subject to:} \quad & x \in P \\ & P = \{x \in \mathbb{R}^n : x^\top d \leq 1, d \in S\}. \end{aligned}$$

Reformulation intuition



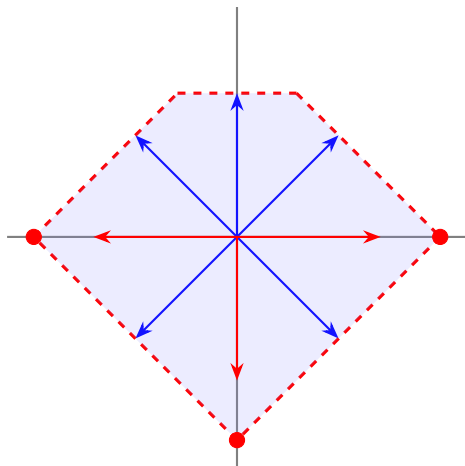
Consider a positive spanning set S .

Reformulation intuition



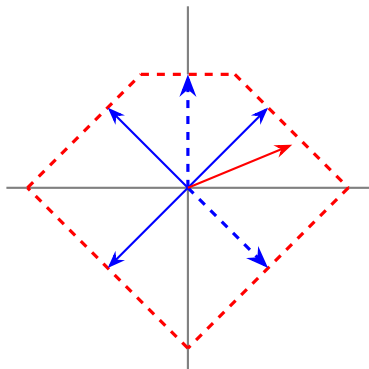
Consider the polytope $P = \{x : x^T d \leq 1, d \in S\}$.

Reformulation intuition

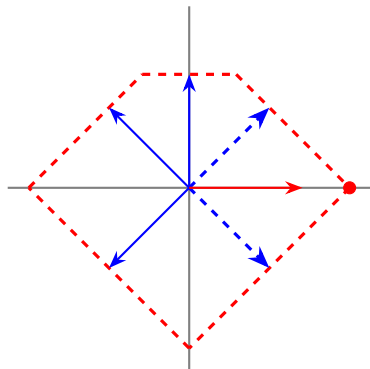


Cosine vectors correspond to “furthest” vertices of P .

Main reformulation theorem



(Invalid vertex)



(Valid vertex)

Main reformulation theorem

Theorem. Let S be a finite positive spanning set and

$$P = \{x \in \mathbb{R}^n : x^\top d \leq 1, d \in S\}.$$

1. Let x be a vertex of P . Then $\frac{x}{\|x\|}$ is a Gram vector of B with Gram value $\frac{1}{\|x\|}$.
2. Let $B \subseteq S$ be a basis and u be the Gram vector of B with Gram value α . If $K^\circ(B, u) \cap S = \emptyset$, then $\frac{u}{\alpha}$ is the unique vertex.

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Summary of current algorithms

1. Enumerate all bases and compute corresponding gram vectors [HJ20]
2. Enumerate all KKT points (also enumerates all bases)[Reg21]

-
- Deterministic
 - Finite runtime
 - Returns the exact solution
 - Expensive in higher dimension: $\binom{\# \text{ of vectors in set}}{n}$ bases

Vertex enumeration

Observation: all solutions lie at a vertex.

Algorithm: Vertex Enumeration Algorithm

Input: Finite positive spanning set S

Enumerate the vertices of $P = \{x : x^\top d \leq 1, \forall d \in S\}$

v^* = vertex of P with largest norm

return

$\frac{1}{\|v^*\|}$: the exact cosine measure

$\left\{ \frac{v}{\|v\|} : v \text{ is a vertex of } P, \|v\| = \|v^*\| \right\}$: the cosine vector set

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- Far fewer vertices than number of subbases

Vertex enumeration

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return

$\frac{1}{\|v^*\|}$: the exact cosine measure

$\left\{ \frac{v}{\|v\|} : v \text{ is a vertex of } P, \|v\| = \|v^*\| \right\}$: the cosine vector set

- Far fewer vertices than number of subbases
- Still an exponential number of vertices

Vertex enumeration

Observation: all solutions lie at a vertex.

Algorithm: Vertex Enumeration Algorithm

Input: Finite positive spanning set S

Enumerate the vertices of $P = \{x : x^\top d \leq 1, \forall d \in S\}$

v^* = vertex of P with largest norm

return

$\frac{1}{\|v^*\|}$: the exact cosine measure

$\left\{ \frac{v}{\|v\|} : v \text{ is a vertex of } P, \|v\| = \|v^*\| \right\}$: the cosine vector set

- Far fewer vertices than number of subbases
- Still an exponential number of vertices
- Returns exact solution in finite time

Random linear programs

$$\max c^\top x \quad \text{subject to} \quad x \in P \quad (1)$$

Algorithm: Random LPs

Input: Finite positive spanning set S , number of iterations k

$v = 0$

for $i = 1, \dots, k$ **do**

 Uniformly sample $c \in \{x : \|x\| = 1\}$

 Let $\bar{v}$ be the solution to LP (1)

if $\|\bar{v}\| > \|v\|$ **then**

$v = \bar{v}$

end

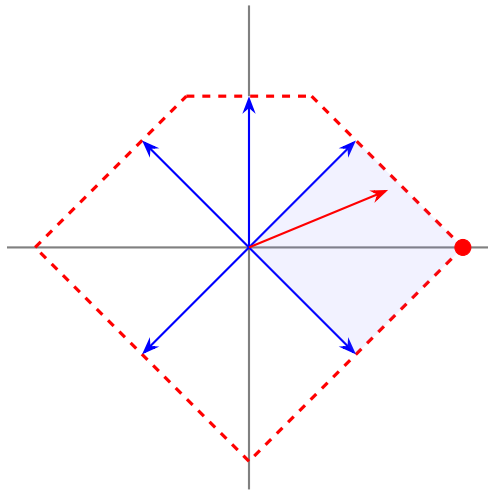
end

return

$\frac{1}{\|v\|}$: an approximation of the cosine measure

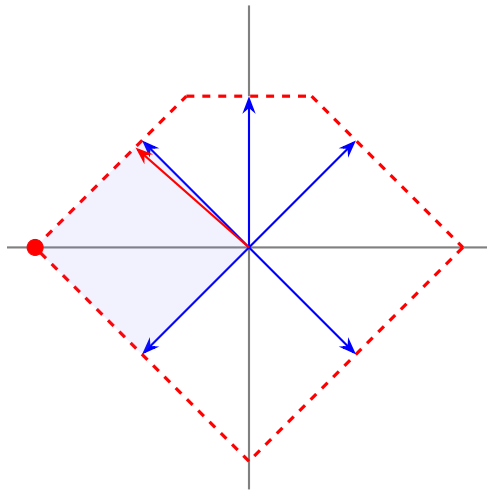
$\frac{v}{\|v\|}$: a vector that provides the approximate cosine measure

Random linear programs



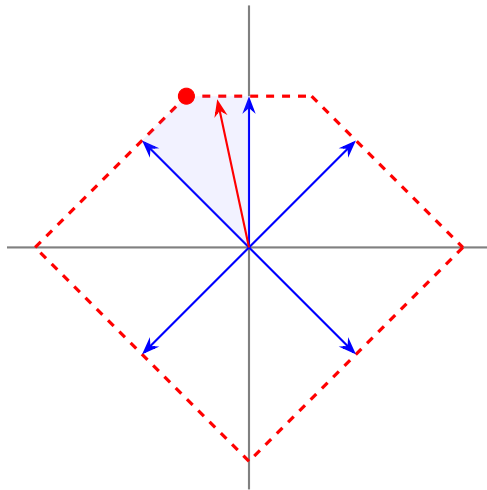
$$\max c^T x \quad \text{s.t.} \quad x \in P$$

Random linear programs



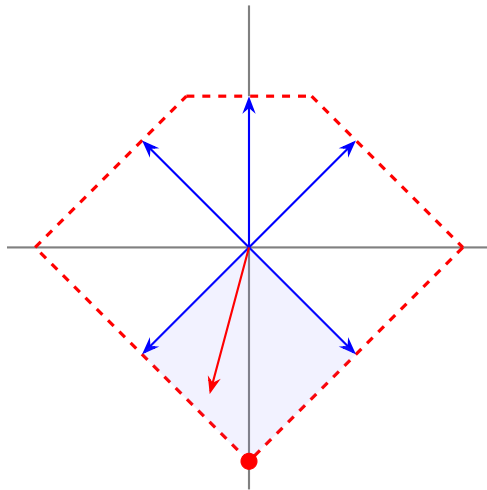
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Random linear programs



$$\max c^T x \quad \text{s.t.} \quad x \in P$$

Random linear programs



$$\max c^T x \quad \text{s.t.} \quad x \in P$$

Random linear programs convergence

Theorem. With probability 1, the random LP method returns a minimizer.

Proof.

- Let v be a vertex of P . Then there exists a basis $B \subset S$ of $\mathbb{R}^n$ such that for all $c \in \text{pspan}(B)$, the solution to the linear program

$$\max c^\top x \quad \text{s.t.} \quad x \in P$$

is v .

- The cone $\text{pspan}(B)$ has non-empty interior, hence has non-zero spherical measure.
- There is a non-zero probability that $c \in \text{pspan}(B)$ when c is drawn uniformly from the unit sphere.



Branch-and-bound

Directly apply branch and bound on the QP formulation:

$$\begin{aligned} & \max_{x \in \mathbb{R}^n} \quad \|x\|^2 \\ & \text{subject to: } x \in P \\ & \quad P = \{x \in \mathbb{R}^n : x^\top d \leq 1, d \in S\} \end{aligned}$$

or the QCLP formulation:

$$\begin{aligned} & \min \quad z \\ & \text{subject to: } d_i^\top v \leq z, \quad v^\top v = 1 \\ & \quad v \in \mathbb{R}^n, \quad z \in \mathbb{R} \end{aligned}$$

Produces an ϵ -optimal global solution if both lower and upper bound close.

Outline

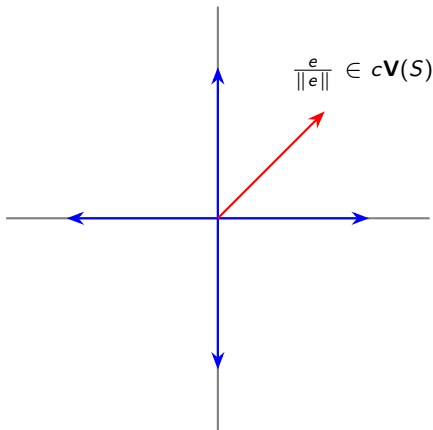
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Sets with known cosine measure

We use the following sets to benchmark the algorithms:

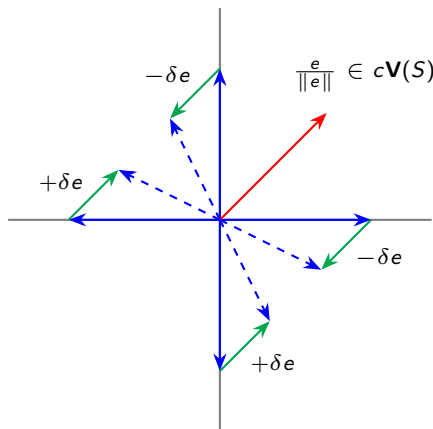
Set Type	Cosine Measure	# of Vectors	Reference
Min. Canonical PB	$\frac{1}{\sqrt{n^2+2(n-1)\sqrt{n}}}$	$n + 1$	[Har+24]
Min. δ -shift PB	$\frac{1-\delta n}{\sqrt{n(\delta^2 n-2\delta+1)}}$	$n + 1$	NEW
Max. δ -shift PB	$\frac{1-\delta n}{\sqrt{n(\delta^2 n-2\delta+1)}}$	$2n$	NEW
Optim. Ortho. PB	$\frac{1}{\sqrt{(s-n-r)\lfloor \frac{n}{s-n} \rfloor^2 + r\lceil \frac{n}{s-n} \rceil^2}}$	$n + 1 < s < 2n$	[HJP23]
Rand. PSS	Unknown	$n + 1 < s < 2n$	[Reg16]

Set with arbitrary cosine measure



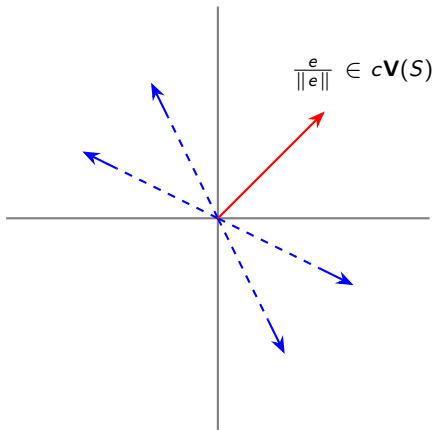
Consider the canonical maximal positive basis. $I \cup -I$

Set with arbitrary cosine measure



Shift each vector in the set by $\pm\delta e$.
Change δ to control the size of the “gap”.

Set with arbitrary cosine measure



Scale the set by α .

Set with arbitrary cosine measure theorem

Theorem. Let $n \geq 2$ and $0 \leq \delta < \frac{1}{n}$. Consider the matrix $B = I - \delta e^{n \times n}$. Define $P^\delta = \alpha B \cup (-\alpha B)$ and $\alpha = \frac{1}{\sqrt{\delta^2 n - 2\delta + 1}}$. Then P^δ is a positive basis with cosine measure $\frac{1 - \delta n}{\sqrt{n(\delta^2 n - 2\delta + 1)}}$.

Proof.

- Show that $\frac{e}{\|e\|}$ is the Gram vector of αB , even after shifting.
- Show that the Gram vector $\frac{e}{\|e\|}$ has the smallest Gram value (candidate cosine measure) by comparing all subbases of P^δ .



Proof sketch

Proof. Let αT be an arbitrary sub-basis of P^δ , then

$$\alpha T = \alpha\{\pm b_1, \dots, \pm b_n\} \subset \alpha B \cup (-\alpha B)$$

so we may write $B = TQ$ for an orthogonal matrix Q . We wish to show

$$\frac{1}{\sqrt{e^\top \mathbf{G}(\alpha B)^{-1} e}} \leq \frac{1}{\sqrt{e^\top \mathbf{G}(\alpha T)^{-1} e}}.$$



Proof sketch

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so we may write $B = TQ$ for an orthogonal matrix Q . We wish to show

$$\frac{1}{\sqrt{e^\top \mathbf{G}(\alpha B)^{-1} e}} \leq \frac{1}{\sqrt{e^\top Q^\top \mathbf{G}(\alpha B)^{-1} Q e}}.$$



Proof. Since

$$\begin{aligned}\mathbf{G}(\alpha B) &= (\alpha B)^\top (B\alpha) \\ &= \alpha^2 (I - \delta e^{n \times n})^\top (I - \delta e^{n \times n}) \\ &= \alpha^2 I + (1 - \alpha^2) e^{n \times n}\end{aligned}$$



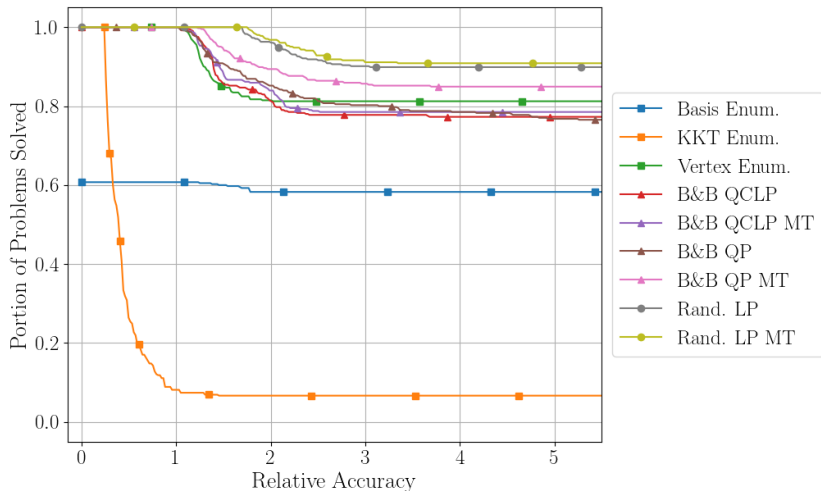
so $\mathbf{G}(\alpha B)^{-1}$ can be obtained using a rank-1 update formula.

A similar idea can be used to construct sets of size $n + 1$ with arbitrary cosine measure.

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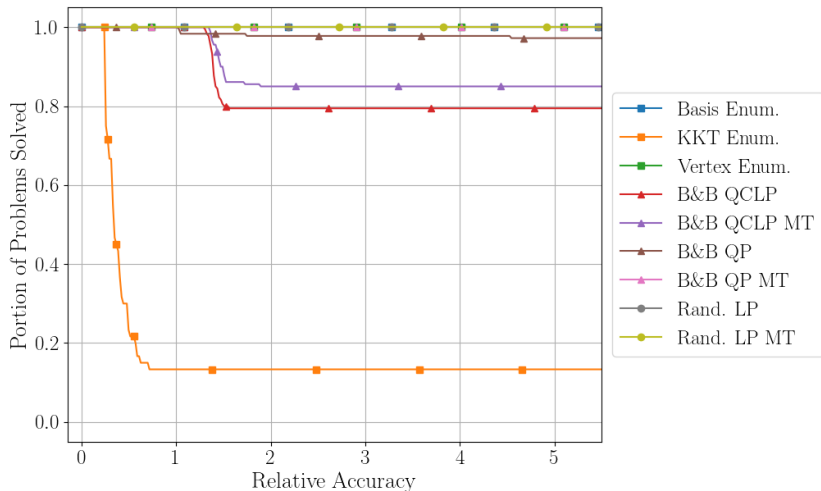
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Results on sets with size $n + 1 \leq |S| \leq 2n$



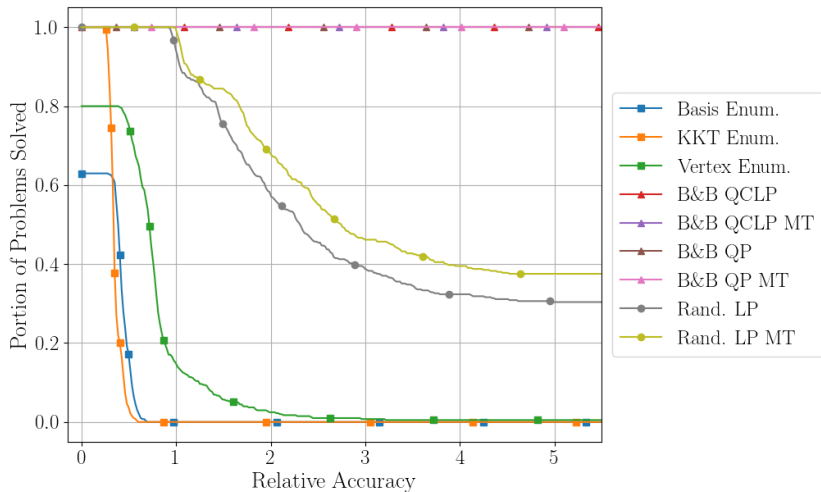
- Random linear program performs well overall.

Results on sets with size $n + 1$



- For sets with $n + 1$ vectors, enumerative methods work well.

Results on sets with a large number of vectors



- Branch-and-bound works perfectly on sets with a large number of vectors.

Conclusion

- Proposed a new reformulation of the cosine measure problem.
- Based on the reformulation, proposed two new heuristics.
- Derived new theorems to generate sets with arbitrary cosine measure.
- Compiled a test collection of sets.
- Produced numerical results comparing the algorithms.

Thank you

Warren Hare and Scholar Sun. *On the computation of the cosine measure in high dimensions*. 2025. arXiv: 2506.19723 [math.OC]. URL: <https://arxiv.org/abs/2506.19723>

